भारतीय प्रौद्योगिकी संस्थान जोधपुर
Indian Institute of Technology Jodhpur

Concept Note for Master of Design (M.Des.) in Smart Product Design

Two-year full-time degree program

School of Design
Indian Institute of Technology Jodhpur

Introduction

The world around us is undergoing a dramatic transformation. We are increasingly interconnected, with a growing number of everyday products evolving into intelligent companions. From self-regulating thermostats to voice-controlled speakers, the physical and digital realms are merging to create a wave of "smart" products. This explosion of smart technology presents both immense opportunities and design challenges.

The Master of Design (M.Des.) in Smart Product Design is crafted to meet the growing demand for professionals who can create innovative, user-centric products that harness the power of smart technology. This program is designed to bridge the gap between cutting-edge technologies, user experience (UX) principles, and design methodology. We aim to delve into the world of smart products, from connected devices to AI-powered systems, learning to design seamless interactions that not only leverage the power of technology but also enrich the lives of people.

This program is ideal for aspiring designers who are passionate about innovation and eager to push the boundaries of physical and digital experiences. As the world embraces smart technology, the demand for skilled designers who can create user-centred, intuitive, and intelligent products is skyrocketing. The M.Des. in Smart Product Design will prepare future designers to be leaders in this dynamic field, shaping the future of how we interact with the world around us.

Background

The smart product design industry is rapidly evolving, driven by advancements in IoT (Internet of Things), AI (Artificial Intelligence), AR (Augmented Reality), VR (Virtual Reality), MR (Mixed Reality), and smart materials. Industries such as consumer electronics, automotive, healthcare, education, and home automation are investing heavily in smart products that enhance user experience through connectivity and intelligence. As technology becomes more embedded in everyday objects, there is a significant need for skilled designers who understand both the aesthetic and functional aspects of smart products. This program is designed to address that need, equipping graduates with the skills to innovate and lead in this transformative field.

The Objectives of the Program

The M.Des. in Smart Product Design aims to cultivate a new breed of design professionals who can:

- Translate user needs and market trends into intelligent product concepts with a comprehensive understanding of the interplay between technology and design.
- Integrate hardware, software, and user interfaces (UIs) seamlessly into product development.
- Apply design methodologies to create user-centered smart products.
- Understand and utilize emerging technologies like AI, machine learning, XR (Extended Reality), and the Internet of Things (IoT) in product design.
- Develop strong communication and collaboration skills to work effectively with cross-functional teams.

This program prepares graduates to lead the design of smart products that are not only technologically advanced but also intuitive, user-friendly, and solve real-world problems.

Expected Graduate Attributes

Graduates of the M.Des. in Smart Product Design will be:

- highly skilled in design methodology, user research, and user-centered design methodologies.
- proficient in design tools for physical and digital prototyping and user interface design for smart products.
- possess a deep understanding of emerging technologies and their application in product design.
- able to think critically and creatively to solve complex design challenges in the smart product domain.
- adept at communicating design ideas effectively to both technical and non-technical audiences.
- prepared to launch successful careers in design consultancies, technology companies, or pursue entrepreneurial ventures in the smart product industry.

Learning Outcomes

Upon successful completion of this program, students will be able to:

- analyze user needs and market trends to identify opportunities for smart product design.

- design, prototype, and test smart products that integrate hardware, software, and user interfaces seamlessly.
- apply design methodologies throughout the entire product development lifecycle.
- develop user-centered design solutions that are intuitive, aesthetically pleasing, and solve user problems.
- collaborate effectively with engineers, developers, and other stakeholders to bring smart products to life.
- articulate the impact of technology on design and the ethical considerations of smart product development.

Program Credit Structure

Course Type	Total Courses	Credits
Compulsory	4	12
Electives	4	12
Non-graded	2	0
Summer Thesis	1	8
Thesis	2 (16+16)	32
Total	**13**	**64**

Semester-wise Distribution of Credits

Semester I						**Semester II**				
Cat	Course Code	Course Title	L-T-P	Credits		Cat	Course Code	Course Title	L-T-P	Credits
C	DSL7680	Design Processes and Methods	2-0-2	3		E	DSL7XXX	Elective	L-T-P	3
C	DSL7360	Interaction Design	2-0-2	3		E	DSL7XXX	Elective	L-T-P	3
C	DSL7520	Studies in Form and Aesthetics	2-0-2	3		E	DSL7XXX	Elective	L-T-P	3
C	DSL7530	Nature of Materials and	2-0-2	3		E	DSL7XXX	Elective	L-T-P	3

		Processes								
NG	LAN7010	Technical Communication	1-0-0	S/U		NG	OAL7010	Innovation and IP Management	1-0-0	S/U
Total				12		Total				12
Summer Term										
T	DSS7700	Summer Thesis for SPD	0-0-8	8						
Toal				8						
Semester III						**Semester IV**				
T	DST7710	SPD Thesis Phase I	0-0-16	16		T	DST7720	SPD Thesis Phase II	0-0-16	16
Toal				16			Total			16

Note: The thesis work must be aligned with the objectives of the enrolled program as stated in the concept note.

List of Electives

Sl. No.	Code	Course	L-T-P	Credits
1	DSL7540	Product Visualisation and Ideation	1-0-4	3
2	DSL7620	IoT Systems for Smart Product Design	2-0-2	3
3	DSL7630	AI and ML for Smart Product Design	1-0-4	3
4	DSL7550	Designing Data Visualisations	1-0-4	3
5	DSL7560	Computer-Aided Design	1-0-4	3
6	DSL7570	Product Semantics	1-0-4	3
7	DSL7580	Nature-inspired Design	1-0-4	3
8	DSL7590	Product Ergonomics	2-0-2	3
9	DSL7600	Advanced CMF Strategies for Smart Product Design	2-0-2	3
10	DSL7610	Special Topics in Smart Product Design	2-0-2	3
11	DSL7690	Computational Design and Digital Fabrication	1-0-4	3
12	DSL7200	Usability Engineering	1-0-4	3

Pedagogy

The program delivery focuses on building a strong theoretical foundation along with rigorous hands-on activities based on real-world design challenges. The curriculum integrates lectures, workshops, collaborative studio projects, and invited sessions by industry experts. Students receive personalised feedback throughout the program to develop their design skills and critical thinking abilities.

Eligibility for Admission

1. A valid CEED scorecard (mandatory only for full-time regulars with assistantship/ fellowship seats).
2. The applicant must have a bachelor's degree (program duration: 4 years or more) **OR** Professional Diploma in Design: 10+2+4 years (minimum 4-year diploma program) **OR** Master's Degree, or an equivalent degree as decided by IIT Jodhpur.
3. A minimum of 60% marks in the qualifying degree for GEN/OBC (55% for SC/ST/PD) category in aggregate or as specified by the university/institute or a minimum CGPA of 6.0 for GEN/OBC (5.5 for SC/ST/PD) category on the scale of 10; with corresponding proportional requirements when the scales are other than on 10 (for example 4.8 for GEN/OBC category, 4.4 for SC/ST/PD on a scale of 8), or a first-class as specified by the university/institute.

Admission Process

Admission to this program will be through the following steps:

1. Submission of online application (for full-time regular with assistantship/fellowship and self-sponsored admission categories)
    i. A valid CEED scorecard (mandatory only for full-time regular with assistantship/fellowship admission category)
    ii. Submission of one or more of the following: Design task, Statement of Purpose (SoP) and Portfolio
2. Shortlisting
    i. Candidates will be shortlisted for interview based on: CEED score (for full-time regular with assistantship/fellowship applications), Design Task, Portfolio, and SoP.
3. Interview of the shortlisted candidates

The final merit list and waiting list for the admission to the program will be prepared based on a comprehensive evaluation of the above elements, including the interview. Admission offers will be made according to the merit list and seat matrix.

Number of Seats

1. Full-time Regular with Assistantship/ Fellowship: 12 seats
2. Full-time Regular Self-Sponsored: 13 seats

Targeted Students and Career Prospects

The program is designed for graduates from design, engineering, architecture, or related disciplines who are passionate about integrating technology with user-centered design. Ideal candidates are those with a curiosity for emerging technologies, a strong sense of empathy, and the desire to create meaningful, interactive, and future-ready products. Graduates can pursue careers as Smart Product Designers, UX Designers for IoT, Interaction Designers, Human-Centered AI Designers, or Product Strategists in sectors such as consumer electronics, healthcare, education, mobility, and home automation. They will also be equipped to take on entrepreneurial roles and pursue doctoral research in related areas.

Industry and Research Collaborations

Strategic partnerships will be established with industries in consumer electronics, healthcare, automotive, and defence sectors to co-develop studio briefs, offer internship opportunities, and sponsor design theses. Research collaborations will be pursued with academic institutions and R&D labs working in AI, IoT, and human-computer interaction. These engagements will ensure that students work on real-world problems and are mentored by professionals, fostering a strong industry-academia ecosystem.

Expected Outcomes and Impact

Graduates of the program will be capable of leading the design of intuitive, sustainable, and intelligent products that integrate hardware, software, and user experience. The program is expected to contribute to national innovation goals by producing designers who are adept at creating smart solutions for healthcare, accessibility, mobility, environmental sustainability, etc. Additionally, it will strengthen the position of the nation in global design innovation by generating intellectual

property, fostering startups, and supporting applied research in emerging design-technology intersections.

Program alignment with the UN Sustainable Development Goals

The Master of Design (M.Des.) in Smart Product Design will align with the United Nations Sustainable Development Goals (SDGs) in multiple ways, ensuring that product innovation contributes to sustainable and responsible development. Below are key alignments with specific SDGs:

- SDG 9 – Industry, Innovation, and Infrastructure

  The program encourages the design of smart, connected products that enhance industrial efficiency and promote sustainable manufacturing. Moreover, the program focuses on IoT, AI, and smart materials to improve product performance while minimising waste.

- SDG 3 – Good Health and Well-being

  The curriculum supports the design of smart healthcare devices, wearables, and assistive technologies for better health monitoring and accessibility. The students will be encouraged to formulate their projects so that the project outcomes are ergonomic and user-friendly and designed to enhance well-being.

- SDG 4 – Quality Education

  The students will be encouraged by hands-on, problem-solving approaches to sustainability through smart product design.

Detailed Course Contents

Title	**Design Processes and Methods**	Number	DSL7680
Department	School of Design	L-T-P/St [C]	2-0-2 [3]
Offered for	M.Des. and Ph.D. (Design)	Type	Core
Prerequisite	None		

Objectives

Introduce the students to

- various design methods that have been developed and adopted by researchers and designers in academia and industry.
- a systematic approach to designing products..

Learning Outcomes

The students will be able to

- apply various methods for problem-solving in a systematic approach.
- select and apply a suitable design process in their design projects as per the project requirement..

Contents

- **Evolution of design processes:** History of design methodology, Design process archetype, Academic models, Consultant models, Cyclic models. *[4 lectures, 2 studio hours]*

- **Study of Designing:** Understanding the three essential stages of designing.
    - **Analysis/Divergence:** Stating Objectives, Literature Searching, Searching for Visual Inconsistencies, Innovation Sourcebook, Initial Opportunity Map, Know Context, Know People, Frame Insights. *[8 lectures, 12 studio hours]*

    - **Synthesis/Transformation:** Explore Concepts, Frame Solutions, Brainstorming, Morphological Charts, Interaction Matrix, Interaction Net. *[8 lectures, 6 studio hours]*

    - **Evaluation/Convergence:** Realize Offerings, Checklists, Selecting Criteria, Ranking, and Weighting. *[6 lectures, 6 studio hours]*

Text books

- Jones, John Chris. Design methods. John Wiley & Sons, 1992.
- Kumar, Vijay. 101 design methods: A structured approach for driving innovation in your organization. John Wiley & Sons, 2012.

References

- Bryan, Lawson. "How Designers Think: the design process demystified." Architectural Press and Elsevier, 2005.
- Ulrich Karl T, Eppinger Steven D.: Product Design and Development, Tata McGraw-Hill, 2004
- James Garrat.: Design and Technology, Cambridge, 1995

Online resources

- Innovation by Design by Prof. B.K. Chakravarthy: https://nptel.ac.in/courses/107101086
- Design, Technology and Innovation by Prof. B.K. Chakravarthy: https://nptel.ac.in/courses/107101088

Course Title	**Interaction Design**	Course No.	DSL7360
Department	School of Design	Structure (L-T-P [C])	2-0-2 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Core
Prerequisite	None		

Objectives: The course aims to:

1. Introduce key principles, models and cognitive foundations of interaction design.
2. Develop skills to design, prototype, and evaluate interactive systems grounded in usability, accessibility, and cognitive principles.
3. Prepare students to design for emerging technologies and multimodal systems, with an awareness of ethical and inclusive practices.

Learning outcomes: By the end of this course, students will be able to:

1. Design and prototype interactive systems that are intuitive, accessible, and user-centered.
2. Analyse user behavior and evaluate interaction designs using appropriate testing methods and usability metrics.

Contents

Foundations of interaction design (5 lectures): Definition, scope, and evolution of interaction design. Key frameworks: Conceptual models, interaction models and UX layers. Cognitive foundations: Mental model, human factors, human limitations. Goals and goal-driven design.

User and context modelling (6 lectures): Design processes: Goal-directed, user-centered activity-centered. Research methods: User needs, defining design requirements. Persona, context mapping, user journeys, task flows, and user flows. System boundaries, and service blueprints.

Designing interactions and interface behavior (6 lectures): Interaction paradigms: WIMP , Post-WIMP, ubiquitous computing. Types of interaction and interfaces. Affordances, feedback, constraints and mappings. Visual perception and layout principles, Navigation models. Accessibility and inclusive design.

Evaluation methods (5 lectures): User experience metrics, Evaluation types: Formative, summative. Methods: Heuristics, cognitive walkthrough, usability testing, A/B testing, analytics. Interpreting data and iterating on design decisions.

Advanced interaction paradigms and ethics (4 lectures): Explores emerging interaction modes. Multimodal interaction: Voice, touch, gesture, gaze, haptics, ambient feedback. Adaptive, XR, spatial, and conversational interfaces. Human-AI collaboration: Transparency, explainability, trust. Affective computing: Emotional and persuasive design. Ethics, Dark patterns, and responsible design.

Studio Exercises (13 sessions)

- Conduct design research to identify user needs, pain points, expectations, and contextual factors
- Develop user personas, journey maps, and task flows to model user behavior and define design requirements
- Generate and explore low-fidelity concepts and wireframing key interactions
- Create high-fidelity interactive prototypes, conduct usability testing, and iterate based on feedback for final deliverables

Text books

1. Sharp, H., Rogers, Y., & Preece, J. (2019). *Interaction design: Beyond human-computer interaction* (5th ed.). Wiley.
2. Norman, D. A. (2013). *The design of everyday things* (Revised and expanded ed.). Basic Books.
3. Cooper, A., Reimann, R., Cronin, D., & Noessel, C. (2014). *About face: The essentials of interaction design* (4th ed.). Wiley.

References

1. Garrett, J. J. (2010). *The elements of user experience: User-centered design for the web and beyond* (2nd ed.). New Riders.
2. Greever, T. (2015). *Articulating design decisions: Communicate with stakeholders, keep your sanity, and deliver the best user experience*. New Riders
3. Moggridge, B. (2007). *Designing interactions*. MIT Press.

Online material

1. NPTEL. (n.d.). *Interaction Design*. Indian Institute of Technology Guwahati. Retrieved March 24, 2025, from https://nptel.ac.in/courses/107103083
2. IDEO.org. (2015). T*he Field Guide to Human-Centered Design*. IDEO.org.

Title	**Studies in Forms and Aesthetics**	Number	DSL7520
Department	School of Design	L-T-P/St [C]	2-0-2 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Core
Prerequisite	None		

Objective: Introduce students to -

- Fundamental principles and elements of influencing form, aesthetics, and UX.
- The relationship between aesthetics, functionality, and technology in SPD.

Learning Outcomes: The students will be able to -

- Apply design principles to create aesthetically pleasing and functional products.
- Integrate form, aesthetics, and usability to enhance UX and product desirability.

Contents

Elements of Design (4 lectures, 6 Studio/Practical hours)
Line, shape, and form: Visual composition and structure in product design; Color theory: Impact of hues, tones, and palettes on perception and emotion; Texture and materiality: Tactile and visual experiences in product aesthetics; Space and balance: Negative space, proportion, and equilibrium in forms.

Principles of Design (6 lectures, 4 Studio/Practical hours)
Harmony and contrast: Achieving visual coherence and intentional disruption; Rhythm and movement: Guiding user interaction through dynamic elements; Emphasis and hierarchy: Directing

attention to key product features; Scale and proportion: Ensuring functionality and ergonomic alignment.

Aesthetics in Smart Product Design (6 lectures, 4 Studio/Practical hours)
Integration of technology and form: Balancing aesthetics with functionality; Minimalism vs. complexity: Adapting design to product context and user needs; Emotional design: Evoking connections through form and aesthetic language; Trends in smart product aesthetics: Future-forward and culturally relevant forms.

Form Exploration and Development (4 lectures, 8 Studio/Practical hours)
Iterative sketching: Techniques for form exploration and refinement; 3D modeling and visualization: Tools for developing and presenting forms; Role of biomimicry in inspiring innovative forms; Material and manufacturing constraints influencing form development.

Evaluation and Critique of Aesthetics (6 lectures, 4 Studio/Practical hours)
User-centered evaluation of form and aesthetics; Aesthetic alignment with brand identity and market positioning; Cultural and societal influences on aesthetic preferences; Refining design through feedback and critique.

References

- Ching, F. D. K. (2015). *Form, space, and order* (4th ed.). Wiley.
- Hekkert, P., & Schifferstein, H. N. J. (Eds.). (2008). *Product experience*. Elsevier.
- Lidwell, W., Holden, K., & Butler, J. (2010). *Universal principles of design* (Revised ed.). Rockport.

Online Resources

- Form and Aesthetics: https://onlinecourses.nptel.ac.in/noc25_de13/preview

Title	**Nature of Materials and Processes**	Number	DSL7530
Department	School of Design	L-T-P/St [C]	2-0-2 [3]
Offered for	M.Des., Ph.D.(Design)	Type	Core
Prerequisite	None		

Objective:
Introduce students to -

- The fundamental properties of materials and their applications in product design.
- Manufacturing processes and their influence on form, function, and sustainability of products.

Learning Outcomes:
The students will be able to -

- Select appropriate materials and processes to optimize product performance and usability.
- Incorporate sustainability and innovation in material and process planning for product development.

Contents

1. Introduction to Materials (4 lectures, 6 Studio/Practical hours)

Classification of materials: Metals, polymers, ceramics, composites, and smart materials; Material properties: Mechanical, thermal, electrical, and optical characteristics; Principles of material selection: Matching properties with design requirements; Material trends: Advanced, bio-based, and sustainable materials.

2. Manufacturing Processes (12 lectures, 10 Studio/Practical hours)
Overview of primary and secondary manufacturing techniques; Plastic forming processes: Injection molding, blow molding, and thermoforming; Metal processing: Casting, forging, and additive manufacturing (3D printing); Process optimization: Balancing cost, quality, and environmental impact.

3. Materials and Processes in Smart Products (5 lectures, 6 Studio/Practical hours)
Integration of sensors and electronics: Materials for connectivity and IoT; Case studies: Successful applications of innovative materials and processes; Material-process compatibility for durability and aesthetics in smart products; Role of prototyping: Testing materials and manufacturing feasibility.

4. Sustainability and Future Trends (5 lectures, 4 Studio/Practical hours)
Eco-design principles: Reducing waste and energy consumption through smart materials; Circular economy: Recycling and reuse in material and process planning; Emerging technologies in manufacturing: AI, robotics, and digital fabrication; Forecasting future developments in materials and processes.

References

1. Ashby, M. F., Shercliff, H., & Cebon, D. (2019). Materials: Engineering, science, processing and design (4th ed.). Elsevier.
2. Crawford, R. J., & Tamar, M. P. (2020). Plastics engineering (4th ed.). Elsevier.
3. Callister, W. D., & Rethwisch, D. G. (2020). Materials science and engineering: An introduction (10th ed.). Wiley.

Online Resources

- Designing of Plastic Products for Injection Moulding: https://www.dsource.in/course/designing-plastic-products-injection-moulding

Fundamentals of Manufacturing Processes: https://onlinecourses.nptel.ac.in/noc23_me131/preview

Title	**Product Visualisation and Ideation**	Number	DSL7540
Department	School of Design	L-T-P [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objective: Introduce students to -

- The advanced techniques in visual thinking and ideation methods for the conceptual development of smart products.
- Sketching and digital visualisation techniques to communicate design ideas effectively.

Learning Outcomes: The students will be able to -

- Apply structured ideation techniques to generate a wide range of design solutions.
- Translate abstract concepts into tangible visual formats using manual and digital tools.

- Communicate design concepts effectively to both technical and non-technical audiences.

Contents
Visual Thinking and Ideation Fundamentals (2 lectures, 12 Studio/Practical hours)
Role of visualisation in the design process; Sketching for thinking vs. sketching for presentation; Divergent and convergent ideation methods (SCAMPER, mind-mapping, morphological analysis)

Manual and Digital Visualisation Techniques (4 lectures, 20 Studio/Practical hours)
Rapid sketching techniques for product and interface ideas; Perspective drawing, viewpoint, light and shadows, depth, colours, material expressions, scale and exploded views; Introduction to tools such as Procreate, Sketchbook Pro, Adobe Illustrator, Fusion 360, Rhinoceros, Solidworks, Keyshot, VRay; Cross-software workflows

Storyboarding and Experience Mapping (3 lectures, 10 Studio/Practical hours)
Visualising user-product interaction scenarios; Constructing use-case journeys and storyboards; Mapping product ecosystems and service blueprints; Creating annotated scenarios with emotional and behavioural cues; Translating invisible interactions (sensors, AI behaviour) into visual narratives; Visual storytelling techniques for design pitches.

Visualisation using Emerging Technology (4 lectures, 10 Studio/Practical hours)
Environment and Space Design for XR experience; Asset creation and optimisation; Creating realistic and hyper-realistic rendering of spaces and environments for applications like Architecture and Interior Design, Healthcare, Industrial, Educational Training, Retail, Gaming, Tourism, Entertainment, Automotive Industries; Using AI-powered tools for product visualisation

References

- Eissen, K., & Steur, R. (2014). Sketching product design presentation.
- Ching, F. D. (2019). Design drawing. John Wiley & Sons.
- Buxton, B. (2010). Sketching user experiences: getting the design right and the right design. Morgan Kaufmann.
- Jennifer Whyte. (2018). Virtual Reality and the Built Environment. Routledge.

Online Resources
- Product Drawing: https://www.dsource.in/course/product-drawing
- Rhinoceros 8: https://www.rhino3d.com/learn/?query=kind:%20jump_start&modal=null

Title	**IoT Systems for Smart Product Design**	Number	DSL7620
Department	School of Design	L-T-P/St [C]	2-0-2 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives
Introduce students to
- the components and architecture of Internet of Things (IoT) systems, focusing on how devices sense, connect, and interact within smart environments.
- the physical and digital behavior of IoT-enabled systems by exploring the role of sensors, actuators, and simple mechatronic systems.
- data flow, connectivity, and system-level thinking in IoT, enabling them to conceptualize and explain the functioning through real-world case studies.

Learning Outcomes
The students will be able to

- explain the structure and function of an IoT system, including how sensors and actuators communicate and interact with users, networks, and digital platforms.
- demonstrate how sensors and actuators influence the behavior of a connected product using simplified system diagrams or flowcharts.
- analyze and communicate how data is generated, transmitted, and utilized in smart products, considering system performance, user context, and constraints.

Contents:

- **Fundamentals of IoT Systems:** What is IoT, History, Evolution, IoT Architectures, Components of IoT Systems, Application of IoT, Case Studies of Smart Product Design (10 Lecture Hours)
- **Sensors, Actuators and Mechatronics Basics:** Introduction to Mechatronics in IoT, Sensors, Actuators, System Behaviour, Practical Exploration (8 Lecture Hours and 13 Studio/Practical Hours)
- **IoT Ecosystem:** Fundamentals of Networking, Data Flow in IoT, IoT Protocols, Cloud Platforms, Holistic View of IoT Systems, Security and Ethics, Future Directions and Opportunities in IoT. (8 Lecture Hours and 13 Studio/Practical Hours)

References:

- Pfister, C. (2011). Getting Started with the Internet of Things: Connecting Sensors and Microcontrollers to the Cloud. Maker Media, Inc.
- Rowland, C., Goodman, E., Charlier, M., Light, A., & Lui, A. (2015). Designing Connected Products: UX for the Consumer Internet of Things. O'Reilly Media, Inc.
- Bahga, A., & Madisetti, V. (2014). Internet of Things: A Hands-on Approach. VPT.

Online Learning Source:

- NPTEL: Introduction to Internet of Things (https://nptel.ac.in/courses/106105166)

Title	**AI and ML for Smart Product Design**	Number	DSL7630
Department	School of Design	L-T-P/St [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives
Introduce students to

- the fundamental concepts of artificial intelligence (AI) and machine learning (ML) with a focus on their roles in enabling smart and intelligent products.
- real-world applications of AI and ML in digital products, including intelligent user interfaces, adaptive systems, recommendation systems, and predictive tools.
- the practical tools, algorithms, and methodologies used to design smart and intelligent products through case studies.

Learning Outcomes

The students will be able to

- explain and differentiate key concepts, models, and techniques in AI and ML, and analyze their applicability in the context of smart product design.
- evaluate and apply AI and ML approaches to real-world smart product scenarios.
- design and propose data-driven solutions using appropriate AI and ML algorithms and tools, and justify their design decisions through critical analysis of case studies and user needs.

Contents

- **Foundations of AI and ML** (8 Lecture Hours, 4 Studio Hours)
  Introduction to Artificial Intelligence and Machine Learning, Types of Machine Learning Algorithms, Basic ML Algorithms, Fundamentals of Data Preprocessing and Feature Engineering (Feature Extraction and Dimensionality Reduction), Introduction to Evaluation Metrics, Introduction to Neural Networks and Deep Learning.

- **Applications of AI and ML in Smart Products** (5 Lecture Hours, 12 Studio Hours)
  Introduction to Computer Vision, Natural Language Processing (NLP) and Internet of Things (IoT), Case Studies on Intelligent User Interfaces, Recommendation Systems for Personalized Experiences, Adaptive Systems and Context-Aware Product Behavior, Predictive Maintenance and Fault Detection, Image and Speech Recognition in Smart Devices, Sensor Data Analysis and Human Activity Recognition, Role of AI and ML in Automating Design Decisions, Ethics and Bias in Smart Products and AI Applications, Use Cases of AI and ML in Wearables, Smart Appliances, and IoT Products, Generative AI and Scope for Product Design.

- **Introduction to Tools and Techniques** (36 Studio Hours)
  Overview of AI and ML Tools (Python, Scikit-Learn, Keras, and TensorFlow), Data Collection, Labeling, and Cleaning, Model Training, Validation, and Hyperparameter Tuning, Using CNNs for Computer Vision-based Smart Products, Deploying AI and ML Models in Resource-Constrained Devices (Edge AI), Evaluating and Interpreting Model Decisions.

References:

- Murphy, K. P. (2012). Machine learning: a probabilistic perspective. MIT press.
- Géron, A. (2022). Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow. " O'Reilly Media, Inc".
- Bishop, C. M. (2006). Pattern recognition and machine learning. Springer.
- James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). An introduction to statistical learning: with applications in R. Springer.
- Shneiderman, Ben (2022). Human-centered AI. Oxford University Press.

Online Learning Sources:

- NPTEL: Introduction to Machine Learning (https://nptel.ac.in/courses/106106139)
- NPTEL: Responsible and Safe AI Systems (https://onlinecourses.nptel.ac.in/noc24_cs132/preview)

Title	**Designing Data Visualizations**	Number	DSL7550
Department	School of Design	L-T-P [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives: The course aims to:

1. Introduce theoretical principles and cognitive foundations of data visualization.
2. Enable students to design effective and inclusive visualizations grounded in perception, interaction design, and storytelling.
3. Develop the ability to think critically and creatively about how data can be represented, interpreted, and communicated through visual design.

Learning outcomes: By the end of this course, students will be able to:

1. Design data visualizations that are clear, usable, and aligned with user needs and data types.
2. Assess the effectiveness of data visualizations in supporting understanding, decision-making, and responsible communication of data.

Contents

Foundations of data visualisation (8 lectures): Definition, scope, and evolution of data visualisation. Roles and purposes. Types of data: Understanding quantitative, categorical, ordinal, and temporal data. Data tasks: Comparison, distribution, relationship, and composition. Visual variables and encoding: Position, length, area, color, shape, and their application in encoding data. Chart types and use: Common chart types, selecting the most appropriate visualisation. Exploratory and explanatory visualizations. Perception, hierarchy and cognitive load in data visualization. Introduction to data visualization tools.

Designing for users and context (8 lectures): Cognitive and perceptual principles in visualization. Managing visual complexity to enhance readability and comprehension. Structuring and organizing information to support clarity, hierarchy, and user navigation. Designing multi-level and comparative views. Integrating visualizations into digital systems. Design strategies for dashboards and information-dense displays. Designing for responsive and adaptive contexts. Accessibility in data visualization.

Interactive data storytelling (6 lectures): Role of interaction in data exploration and comprehension. Interaction techniques: filtering, zooming, panning, etc. Visual storytelling: Framing, sequencing, and annotation. Narrative structures of data: Temporal, hierarchical, spatial, comparative, and anecdotal narrative structures. Storytelling for insight, advocacy, decision support.

Evaluation, ethics, and future trends (4 lectures): Evaluation techniques for data visualizations. Measuring impact. Bias in data and design. Misleading visualizations. Ethical principles in data representations. Emotional impact and visual rhetoric. Emerging trends in data visualisation.

Studio Exercises (13 sessions)

- Explore and analyze datasets to identify key insights and user needs
- Develop low- and high-fidelity visualizations to communicate data effectively
- Use visualization tools to create interactive charts, maps, and dashboards
- Conduct user feedback sessions to evaluate clarity and usability, and refine designs

Text books

1. Cairo, A. (2012). The functional art: An introduction to information graphics and visualization. New Riders.
2. Tufte, E. R. (2001). The Visual Display of Quantitative Information (2nd ed.). Graphics Press.
3. Munzner, T. (2014). Visualization Analysis and Design. CRC Press.

References

1. Healy, K. (2018). Data Visualization: A Practical Introduction. Princeton University Press.
2. Ware, C. (2020). Information Visualization: Perception for Design (4th ed.). Morgan Kaufmann.
3. Steele, J., & Iliinsky, N. (Eds.). (2010). Beautiful visualization: Looking at data through the eyes of experts. O'Reilly Media.

Online material

1. Massachusetts Institute of Technology. (2017). Visualization [Course module]. In 15.071 The Analytics Edge. MIT OpenCourseWare. https://ocw.mit.edu/courses/15-071-the-analytics-edge-spring-2017/pages/visualization
2. Healy, K. (2017). Data visualization: A practical introduction. https://socviz.co

Title	**Computer-Aided Design**	Number	DSL7560
Department	School of Design	L-T-P [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objective: Introduce students to -

- The industry-standard CAD tools and digital modelling workflows for the design and development of smart products
- The parametric modelling, solid/surface modelling, rapid iteration, digital-physical integration, and preparation for prototyping and manufacturing.

Learning Outcomes: The students will be able to -

- Develop 3D models of smart products using parametric and surface modelling tools.
- Apply CAD techniques for visualisation, prototyping, simulation, and fabrication.
- Create assemblies and design components with functional, ergonomic, and technical considerations.
- Prepare documentation and export formats for 3D printing and fabrication.

Contents

Introduction to CAD for Smart Product Design (2 lectures, 12 Studio/Practical hours)
Overview of CAD applications in design engineering and manufacturing; Understanding CAD environments such as solid, surface, mesh, and hybrid modelling; Comparative introduction to tools: Fusion 360, SolidWorks, Rhino, Blender

Parametric Modelling and Feature-Based Design (4 lectures, 14 Studio/Practical hours)
Sketch-based modelling and geometric constraints; Measurement and mapping; Creating complex parts through extrusions, lofts, and sweeps; Construction planes, patterns, mirroring, and feature

reuse for iterative design; Assembling multi-part products with joint types and degrees of freedom; Managing relationships between components and subassemblies; Introduction to motion simulation for interactive smart products

Surface Modelling and Aesthetic Form Exploration (4 lectures, 14 Studio/Practical hours)
Freeform design using splines and NURBS surfaces; Blending and sculpting techniques for CMF-driven forms; Reverse engineering and mesh-to-CAD workflows

Design for Manufacturing and Prototyping (3 lectures, 12 Studio/Practical hours)
Design for additive and subtractive processes; Tolerance analysis, draft angles, and material properties; Generative Design and Topology Optimization; File preparation and export for 3D printing, laser cutting, and CNC machining; Creating technical drawings and exploded views; Applying materials, lighting, and cameras for realistic rendering; Annotating and exporting design documentation for collaboration and presentation

References

- Rogers, K. C. (2012). Parametric Modeling with Autodesk Fusion 360. SDC Publications
- Shih, R. (2025). Learning SOLIDWORKS 2025: Modeling, Assembly and Analysis. SDC Publications.
- Fisher, G. (2014). Blender 3D Basics Beginner's Guide: A quick and easy-to-use guide to create 3D modeling and animation using Blender 2.7. Packt Publishing Ltd.

Online Resources

- Autodesk Design Academy: https://academy.autodesk.com
- Dassault Systèmes SolidWorks Tutorials: https://www.solidworks.com/sw/resources/getting-started.htm

Title	**Product Semantics**	Number	DSL7570
Department	School of Design	L-T-P/St [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives: Introduce students to:

- The theoretical foundations of product semantics and its role in design.
- How products communicate meaning, values, and intentions to users.

Learning Outcomes: The students will be able to:

- Identify and interpret the semantic dimensions of products.
- Apply principles of product semantics to enhance user understanding and interaction.

Contents

- **Introduction to Product Semantics:** Definition and historical context of product semantics; Relationship between form, function, and meaning; Semiotics in design (signs, symbols, icons); Importance of product language in user experience. [2 lectures, 4 studio/practical

hours]

- **Theories of Meaning and Interpretation:** Cognitive semantics and mental models; Cultural context and cross-cultural communication in design; Affordances and perceived usability; Emotional responses to product form and meaning. Examine how meaning is constructed and perceived in non-physical products, including screen-based and interactive experiences. [3 lectures, 8 studio/practical hours]

- **Designing for Communication:** Expressive qualities of materials, colors, and textures; Typology and archetypes in product design; Narrative and storytelling through products; Designing for specific user groups and contexts. [3 lectures, 12 studio/practical hours]

- **Methods for Semantic Analysis:** Qualitative and quantitative methods for studying product meaning; User testing for semantic interpretation; Case studies of products with strong or ambiguous semantic messages; Deconstructing product language. [2 lectures, 12 studio/practical hours]

- **Product Semantics in Practice:** Applying semantic principles in smart product design, inclusive design, deceptive design, and dark patterns; Ethical considerations in product communication (e.g., deceptive design); Future trends in product language and human-product interaction. [3 lectures, 16 studio/practical hours]

References

- Krippendorff, K. (2006). *The Semantic Turn: A New Foundation for Design*. CRC Press.
- Norman, D. A. (2004). *Emotional Design: Why We Love (or Hate) Everyday Things*. Basic Books.
- Vihma, S. (1995). *Products as Representations*. University of Art and Design Helsinki.

Title	**Nature-Inspired Design**	Number	DSL7580
Department	School of Design	L-T-P/St [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives: Introduce students to:

- The core principles and methodologies of nature-inspired design (biomimicry).
- How biological systems and natural processes can inform innovative design solutions.

Learning Outcomes: The students will be able to:

- Identify and analyze biological strategies for solving design challenges.
- Apply biomimicry principles to conceptualize and develop novel design solutions.

Contents

- **Introduction to Nature Inspired Design:** Definition and history of biomimicry; Levels of biomimicry (form, process, ecosystem); Why look to nature for design; Case studies of iconic nature-inspired designs. [2 lectures]

- **Biological Principles and Design Strategies:** Understanding natural selection and adaptation; Principles of efficiency, resilience, and regeneration in nature; Learning from natural materials, structures, and processes (e.g., self-assembly, hierarchical structures). [2 lectures, 16 studio/practical hours]

- **Biomimicry Methodologies and Tools:** The "Ask Nature" approach; Bio-inspired design process (identify, biological solution, abstract, apply, evaluate); Tools for biological research and translation; Cross-disciplinary collaboration. [3 lectures, 16 studio/practical hours]

- **Case Studies and Applications:** Nature-inspired design in product design, architecture, engineering, and systems design; Examples from diverse fields (e.g., transportation, energy, medicine, materials science); Exploring local biodiversity for inspiration. Potential role in addressing global challenges (e.g., climate change, resource scarcity) and urban planning, sustainability modelling [4 lectures, 20 studio/practical hours]

- **Ethical Considerations and Sustainable Impact:** The role of nature-inspired design in circular economy and sustainability; Ethical sourcing and impact assessment; Future trends in biomimicry and regenerative design; Designing with nature, not just from nature. [2 lectures]

References

- Benyus, J. M. (2002). *Biomimicry: Innovation inspired by nature*. Harper Perennial.
- Pawlyn, M. (2016). *Biomimicry in architecture* (2nd ed.). RIBA Publishing.
- Baumeister, D. (2014). *Biomimicry Resource Handbook*. Biomimicry 3.8.
- NPTEL course - Introduction to Biomimicry https://onlinecourses.nptel.ac.in/noc22_ge24/preview

Title	**Product Ergonomics**	Number	DSL7590
Department	School of Design	L-T-P/St [C]	2-0-2 [3]
Offered for	M.Des., PhD. (Design)	Type	Elective
Prerequisite	None		

Objective: Introduce students to -
- The principles of ergonomics affecting product design and usability.
- Techniques for designing products that enhance user comfort, safety, & efficiency.

Learning Outcomes: The students will be able to -
- Apply ergonomic principles to create user-centered smart products that meet physical and cognitive needs.
- Evaluate and optimize product designs for usability, comfort, and safety through ergonomic testing.

Contents

Fundamentals of Ergonomics (8 lectures, 6 Studio/Practical hours)
Definition and scope of ergonomics in product design; Human factors: Physical, cognitive, and organizational aspects; Anthropometry: Body dimensions and their applications in design; Ergonomic design goals: Comfort, efficiency, and safety.

Physical Ergonomics (6 lectures, 8 Studio/Practical hours)
Designing for posture and movement: Chairs, handles, and wearable devices; Force and interaction: Controls, grips, and haptic feedback in smart products; Accessibility in design: Inclusive solutions for diverse user groups; Ergonomics in smart devices: Adjustability, portability, and user-friendly interfaces.

Cognitive Ergonomics (8 lectures, 6 Studio/Practical hours)
User perception and mental workload in smart product interactions; Information design: Display, feedback, and usability in digital interfaces; Decision-making and error prevention through intuitive design; Emotional ergonomics: Designing for engagement and positive user experiences.

Ergonomics Testing and Evaluation (4 lectures, 6 Studio/Practical hours)
Usability testing methods: Observational studies, surveys, and task analysis; Tools for ergonomic evaluation: Simulations, mock-ups, and virtual reality; Iterative design: Refining products based on user feedback and ergonomic findings; Case studies: Ergonomic successes and failures in smart product design.

References

- Bridger, R. S. (2017). *Introduction to human factors and ergonomics* (4th ed.). CRC Press.
- Kroemer, K. H. E., Kroemer, H. B., & Kroemer-Elbert, K. E. (2001). *Ergonomics: How to design for ease and efficiency* (2nd ed.). Prentice Hall.
- Sanders, M. S., & McCormick, E. J. (1993). *Human factors in engineering and design* (7th ed.). McGraw-Hill.

Online Resources
- Ergonomics for beginners: Industrial design perspective: https://nptel.ac.in/courses/107103004
- Applied Ergonomics: https://onlinecourses.nptel.ac.in/noc20_me11/preview

Title	**Advanced CMF Strategies for Smart Product Design**	Number	DSL7600
Department	School of Design	L-T-P [C]	2-0-2 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objective: Introduce students to -
- The strategies in Colour, Material, and Finish (CMF) with a specific focus on smart products.
- The user context in the CMF development process.
- Influence of CMF on product perception, usability, and interaction.

Learning Outcomes: The students will be able to -

- Develop advanced CMF strategies that align with brand identity, technological function, and user interaction.
- Integrate emerging materials and intelligent surfaces into smart product designs.
- Evaluate the aesthetic, tactile, and functional impact of CMF choices in connected products.
- Communicate CMF intent through detailed specification boards and digital tools

Contents

Foundations and Futures of CMF (4 lectures, 6 Studio/Practical hours)
CMF in product semantics and perception; Historical context and emerging trends; CMF strategy in smart product ecosystems; Trend analysis and forecasting; Colour psychology and emotional resonance; Functional use of colour - feedback, status, guidance; Colour dynamics in illuminated and digital interfaces

Surface Finishes and Manufacturing Constraints (8 lectures, 10 Studio/Practical hours)
Surface treatments: coatings, textures, and printing techniques; Tactile interaction and haptic feedback via finishes; Scalable CMF solutions for mass/custom production;

CMF Specification and Communication (7 lectures, 5 Studio/Practical hours)
CMF boards and storytelling; Digital tools for CMF rendering and documentation (e.g., KeyShot, Adobe Substance, Pantone Studio); translating CMF into manufacturable specs

Sustainability and Culture in CMF (7 lectures, 5 Studio/Practical hours)
Life-cycle thinking in CMF strategy; Local materials and cultural contexts; Responsible sourcing, recycling, and disassembly considerations

References

- Karana, E., Barati, B., Rognoli, V., & van der Laan, A. Z. (2020). Material Experiences: Fundamentals of Materials and Design. Butterworth-Heinemann.
- Ashby, M. F., & Johnson, K. (2013). Materials and design: the art and science of material selection in product design. Butterworth-Heinemann.
- Becerra, L. (2016). CMF design: The fundamental principles of colour, material and finish design. Frame Publishers.

Online Resources

- Material ConneXion Database – https://www.materialconnexion.com
- Pantone Color Institute – https://www.pantone.com

Title	**Special Topics in Smart Product Design**	Number	DSL7610
Department	School of Design	L-T-P [C]	2-0-2 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives
1. Introduce the students to the latest upcoming fields in the area of Smart Product Design (SPD)

Learning Outcomes
1. Apply the knowledge of recent topics to specific research areas in the field of Smart Product

Design (SPD)

Contents
The topic cloud for the course includes contemporary topics in Smart Product Design (SPD) and may be updated according to the instructor.

Text Book
Relevant Textbook and/or research papers to be announced by the instructor.

Self Learning Material
Relevant Textbook and/or research papers to be announced by the instructor.

Title	Computational Design and Digital Fabrication	Number	DSL7690
Department	School of Design	L-T-P [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives
The instructor will introduce students to -
- The computational design methodology encompassing algorithmic logic, data-driven form generation, and design-to-fabrication reasoning
- The digital fabrication using 3D printing and laser cutting

Learning Outcomes
The students are expected to have the ability to:
- Apply computational and parametric thinking as a design methodology, with awareness of its translation into digital fabrication
- Operate 3D visual programming environment proficiently
- Prepare parametric geometry for digital fabrication and develop a portfolio-ready parametric product that bridges digital design and physical output

Contents
Parametric & Computational Design Foundations (4 lectures, 18 Studio/Practical hours)
Introduction to computational thinking in product design: algorithms, parameters, and design logic, Design-to-fabrication workflow: from parametric model to physical output; Grasshopper 3D interface: canvas, component library, panel organisation, and Rhino viewport integration; Understanding inputs, outputs, and data flow; display modes and baking geometry

Data Structures & Advanced Logic (5 lectures, 18 Studio/Practical Sessions)
Logic operators, boolean data, and conditional statements in parametric systems; Primitive curve types: lines, arcs, circles, NURBS curves - parametric construction and control; Primitive surfaces and polysurfaces: planes, spheres, cylinders, lofts - generation and parameter access; Data in Grasshopper: items, lists, and data trees; graft, flatten, and path mapper concepts; resolving input mismatches; cross-referencing strategies; Attractor logic: point, curve, and field-based variation; Graph Mapper and Image Sampler principles

Digital Fabrication (4 lectures, 16 Studio/Practical Sessions)

3D printing workflow: slicing, support generation, orientation strategies, print and review; CNC/Laser fabrication: generating 2D drawings and toolpaths from Grasshopper geometry; manipulating G-Code, DXF export, nesting, and material setup

Laboratory Experiments
Constructing Geometric Hierarchy, Parametric Curve Generation and Control, Lofting, Sweeping, and Ruled Surfaces, Solid Operations for Product Geometry

Textbook
- Kim, E.M., Rhee, J., (2020), Digital Media Series: Grasshopper,
- Marco, G., (2019), Simplified Complexity
- Tedeschi, G. B. A. Aad Algorithms-Aided Design. Parametric Strategies Using Grasshopper

Reference Books
1. García Cuevas, D., & Pugliese, G. (2020). Advanced 3D printing with grasshopper: clay and FDM

Online Course Material
1. Parametric House - Grasshopper tutorials, definitions and scripts - parametrichouse.com
2. Thingiverse - Mesh reference models for 3D printing - thingiverse.com
3. Make: Magazine Digital Fabrication guides - makezine.com

Title	**Usability Engineering**	Number	DSL7200
Department	School of Design	L-T-P/St [C]	1-0-4 [3]
Offered for	M.Des., Ph.D. (Design)	Type	Elective
Prerequisite	None		

Objectives
Introduce the students to
- The importance of the usability of a digital product.
- The methods and techniques for evaluating the usability.

Learning Outcomes
The students will be able to
- Understand the importance of usability in the context of interactive products.
- Analyse user needs and requirements for designing products that are both useful and usable.
- Gain hands-on experience in rapid prototyping and evaluating the usability of interactive products.

Contents

Module 1: [4 lectures, 15 hours of studio]
- Introduction to usability engineering and its significance in interaction design
- Introduction to user experience design and methodologies
- Elements that contribute to user experience
- Establishing user experience goals

Module 2: [6 lectures, 15 hours of studio]
- The process of user-centred design

- User requirement gathering, data collection and analysis
- Core principles, heuristics, guidelines and standards of human-computer interaction
- Usability attributes of immersive digital products

Module 3: [4 lectures, 26 hours of studio]
- Applying gathered and analysed data in the user-centred design process
- Conceptualization and rapid prototyping techniques
- Heuristic and usability evaluation
- Introduction to qualitative and quantitative evaluation techniques

References:

- Dumas, J. S., Dumas, J. S., & Redish, J. (1999). A practical guide to usability testing. Intellect books.
- Nielsen, J. (1994). Usability engineering. Morgan Kaufmann.
- Dix, A., Finlay, J., Abowd, G. D., and Beale, R. (2004). Human-computer interaction. Pearson Education.
- Holtzblatt, K., & Beyer, H. (1997). Contextual design: defining customer-centered systems. Elsevier Morgan Kaufmann.
- Holtzblatt, K., Wendell, J. B., & Wood, S. (2004). Rapid contextual design: a how-to guide to key techniques for user-centered design. Elsevier Morgan Kaufmann
- Goodman, E., Kuniavsky, M., & Moed, A. (2012). Observing the user experience: A practitioner's guide to user research. Elsevier Morgan Kaufmann.
- Rogers, Y., Sharp, H., Preece, J. (2019) Interaction Design: Beyond Human-Computer Interaction. Wiley.

Title	**Summer Thesis**	Number	DSS7700
Department	School of Design	L-T-P/St [C]	0-0-8 [8]
Offered for	M.Des. SPD	Type	Summer Thesis
Prerequisite	None		

Objectives

The Instructor will:
- Introduce students to the conventions of design research
- Guide students in identifying a researchable problem in the field of smart product design
- Provide structured feedback on problem framing, research questions, and preliminary design directions

Learning Outcomes

The students are expected to have the ability to:
- Articulate a well-scoped design research problem grounded in evidence and program-relevant themes
- Conduct a focused literature and precedent review that identifies knowledge gaps and design opportunities.
- Apply appropriate design research methods to gather primary insights and formulate preliminary design hypotheses, research questions, and a conceptual framework

Contents

Introduction to design research, Problem identification, domain mapping, and stakeholder analysis, Literature review, Research documentation

Note: The thesis work must be aligned with the objectives of the enrolled program as stated in the concept note.

Title	**SPD Thesis Phase I**	Number	DST7710
Department	School of Design	L-T-P/St [C]	0-0-16 [16]
Offered for	M.Des. SPD	Type	Thesis
Prerequisite	None		

Objectives
The Instructor will:

- Guide students in translating research insights from the Summer Thesis into structured design opportunities and concept directions
- Facilitate systematic ideation, concept generation, and design exploration through studio-based critique
- Enable students to develop and evaluate multiple design concepts against user, technical, and contextual criteria

Learning Outcomes
The students are expected to have the ability to:

- Synthesise research findings into actionable design directions
- Generate a diverse set of design concepts and evaluate and select a concept direction based on criteria that encompass user desirability, technical feasibility, and program relevance
- Develop low- to mid-fidelity prototypes to explore form, interaction, and functional logic of the proposed smart product

Contents
Insight synthesis and opportunity framing, Concept generation using techniques like brainstorming, sketching, morphological charts, and scenario-based ideation, Design brief development, problem statement, user personas, use scenarios, and design criteria, Concept development and iterative refinement, competitive benchmarking, Low to mid-fidelity prototyping using paper models, digital mockups, breadboarding, and form models

Note: The thesis work must be aligned with the objectives of the enrolled program as stated in the concept note.

Title	**SPD Thesis Phase II**	Number	DST7720
Department	School of Design	L-T-P/St [C]	0-0-16 [16]
Offered for	M.Des. SPD	Type	Thesis
Prerequisite	None		

Objectives
The Instructor will:

- Support rigorous user testing, design iteration, and evidence-based refinement of the thesis outcome
- Oversee the production of a comprehensive thesis document meeting scholarly and professional standards
- Evaluate the thesis work against program objectives, research contributions, and design quality

Learning Outcomes
The students are expected to have the ability to:

- Develop a high-fidelity prototype or design outcome that demonstrates technical resolution, aesthetic coherence, and user-centred rigour
- Plan and execute structured user evaluations, usability tests, or expert appraisals, and integrate findings into design decisions
- Produce a complete thesis document and research publication that articulate the research problem, methodology, design process, outcomes, and reflective critique

Contents
Design development: detailed design, engineering integration, material selection, and manufacturing considerations, User evaluation: usability testing protocols, think-aloud studies, expert reviews, and heuristic evaluation, Iteration and refinement: design changes driven by evaluation evidence and documented rationale, Final prototype or artefact production: high-fidelity model, working system, or equivalent resolved outcome, Thesis writing and research publication writing: abstract, introduction, literature review, methodology, design process, outcomes, and conclusions

Note: The thesis work must be aligned with the objectives of the enrolled program as stated in the concept note.

NAC Feedback

Sr. No.	Name of Expert and Affiliation	Comments on the Concept Note
1	Prof. Anindya Deb, IISc Bangalore	The proposed program is extremely relevant for supporting demanding manpower requirements in the country's foray towards achieving self-reliance and dominance in the manufacturing of complex systems through advanced product design

2	Prof. Lalit Kumar Das, IIT Delhi (Retired)	This initiative represents a timely and significant advancement in both professional design practice and design pedagogy. Having reviewed the program outline, I find it well-conceived and ready for launch in August 2025 as planned.
3	Prof. Braj Bhushan, DD, IITK	At the outset, I compliment team IIT Jodhpur for launching MDes program in Smart Product Design. The core courses suggested by you, Design Thinking, Interaction Design, Form and Aesthetics, and Nature of Materials and Processes, are apt. The basket of electives also looks fine. However, you can consider the following courses in the near future— Design Process & Methodology, Affordable & Sustainable design, User Experience Design, AI in Product Design, and Tangible Interface Design. Additionally, a course or workshop on Innovation Strategy and Product Marketing Planning will also be of immense use to students.
4	Prof. P. V. Madhusudhan Rao, IIT Delhi	The new programme is welcome, much needed and congratulations for starting this.
5	Mr. Ashish Gupta, Director, Antifragile Design LLP, Bengaluru	Discussed and Comments implemented.
6	Mr. Sukanta Maharana, Sr. Industrial Designer, TATA Elxsi Limited, Bangalore	Discussed and Comments implemented.
7	Mr. Chandrashekhar Vyavahare, Director, Futuring Design Studio, Pune	Discussed and Comments implemented.